

VLA-JEPA - browser rehearsal guide

Noah Sprenger and Fernando Nogueira

The measured two-voice narration runs **8:45**. Noah speaks for 4:22 on scenes 1-8; Fernando speaks for 4:22 on scenes 9-18. Planned silence and changes total about 20 seconds. There is no padding to fill a fixed slide budget.

Browser controls

Arrow keys change scenes. Space plays or pauses the independent motion sequence. Build advances the independent local motion. End or Final view shows the complete scene. Home replays. The chapter rail navigates the whole talk. N toggles notes. Navigation and playback controls appear near the screen edges. Narration continues through the main talk; local motion has a separate clock.

The exported movie is the automatic, narrated version. Browser motion follows the available display refresh rate. The movie export uses constant 60 fps at 1920 x 1080. Hardware and browser conditions can limit interactive frame rate.

The light theme, custom scenes, real example frames and continuous flow curve replace the original full-slide images. The opening and conclusion use a procedural Three.js feature sculpture. The method uses actual SSV2 video frames. Feature ribbons and flow trajectories are explanatory illustrations, not measured model outputs. Shared objects interpolate their positions between method chapters. Benchmark values morph in the same chart stage, and both comparators move together. The robot example displays recorded Cartesian pose commands, normalized per axis. B1-B4 are optional references outside the main narrated talk.

Timing and validation

Scene	Speaker	Start	Length
1	Noah	0:00	23.0s
2	Noah	0:22	37.9s
3	Noah	1:00	43.8s
4	Noah	1:44	45.2s
5	Noah	2:29	23.0s
6	Noah	2:52	23.0s
7	Noah	3:15	25.9s
8	Noah	3:41	41.0s
9	Fernando	4:22	35.3s
10	Fernando	4:58	29.4s
11	Fernando	5:27	35.9s
12	Fernando	6:03	14.9s
13	Fernando	6:18	24.8s
14	Fernando	6:43	21.1s
15	Fernando	7:04	19.2s
16	Fernando	7:23	34.3s
17	Fernando	7:57	28.5s
18	Fernando	8:26	19.4s

Content check

The original speech and the three revised segments were transcribed and checked against the intended dialogue. All final scenes and selected intermediate build states were rendered for inspection. These checks establish timing, consistency, and visual fit; they do not measure either presenter's delivery or the audience's understanding. A spoken rehearsal and a classmate comprehension check are still useful.

Analytical flow check

The first half establishes the data gap, related work, predictive supervision, and the information boundary. The second half explains action generation, distinguishes the two training sources, then separates benchmark comparisons from the human-video ablation and preliminary deployment findings. Scenes 4-7 serve different roles: overview, encoding, prediction, and alignment. Implementation settings are kept in notes rather than presented as the audience's main takeaway.

Check audience understanding

After rehearsal, ask a classmate: What does human video supervise? Why are latent tokens not motor commands? How do flow-matching training and inference differ? Which evidence supports robustness, and which counterexample limits the claim? A correct answer should separate future targets from policy inputs and paper results from our preliminary quantization pilot.

Course requirements

The narrated movie is under the ten-minute limit and focuses on the method. The deck includes related work and critical result analysis. Upload the final video to the course Teams location and the PDF to the assignment location. Actual timely participation in course discussion cannot be demonstrated by these artifacts.

Sources

VLA-JEPA paper: <https://arxiv.org/abs/2602.10098v1>

Human footage and its three key frames: Something-Something-v2 validation sample 174198, putting jar into box. Retrieved from the public morpheushoc/something-something-v2 archive mirror and checked against validation.json. <https://www.qualcomm.com/developer/software/something-something-v-2-dataset> ; <https://huggingface.co/datasets/morpheushoc/something-something-v2>

Robot video on slide 2: DROID raw episode AUTOLab+0d4edc83+2023-10-27-19h-52m-50s, exterior camera 24400334. Timing restored from recorded control timestamps; muted, cropped, with an end hold. <https://droid-dataset.github.io/visualizer/>

Related work: UniVLA, <https://arxiv.org/abs/2505.06111> ; LAPA, <https://latentactionpretraining.github.io/> .

UniVLA already learns in DINO feature space; the main comparison distinguishes pipelines and the information entering the latent-action pathway.

The precise cue times below are practical speaking checkpoints estimated across each audio segment. They are not word-level forced alignment or experimentally optimal pause timings. The voices are synthetic stock voices, Puck and Charon, generated using google/gemini-3.8-flash-tts.

Scene 1 - Noah

VLA-JEPA

Clock 0:00-0:22; measured allocation 23.0 seconds.

0:00 - Introduce the speakers warmly, then place gentle emphasis on the question.

0:11 - Let the central question appear; use a slight pitch rise, then settle.

Dialogue

Hi everyone. We're Noah and Fernando, and today we're looking at VLA JEPA. The idea starts with a simple question: if a robot watches someone move an object, what should it actually learn from that video? We'll explain how the paper turns that question into a training method, and then look at where the evidence is convincing and where it still falls short.

Sun et al., VLA-JEPA, arXiv:2602.10098v1 (2026), abstract and introduction.

Scene 2 - Noah

What can video teach a robot?

Clock 0:22-1:00; measured allocation 37.9 seconds.

0:00 - Ask the opening question. Let both real clips play; keep the cursor still.

0:06 - The human clip resolves into before, during and after frames. Both clips now hold.

0:10 - Explain the useful transition before the data gap.

0:24 - Stress “control commands” versus “observable transitions.”

Dialogue

Imagine asking a robot to put an apple into a bowl. Just watch the movement for a moment. What information would help the robot do that itself?

The useful part is the transition: where the object starts, how it moves, and where it ends up. But a video also contains things we don't want the robot to depend on, like the background or the camera angle. Robot demonstrations give us actual control commands, but collecting them takes work. Human videos are much easier to find; they just don't tell us which commands a robot should execute. That's the gap this paper tries to bridge.

Human footage: Something-Something-v2 validation sample 174198.webm, label putting jar into box, template Putting [something] into [something]. Retrieved from the public morpheushoc/something-something-v2 archive mirror and checked against its validation.json annotation; native 427x240, 12 fps, 5.916 seconds. Cropped to 426x192; output duplicates source frames at 60 fps. Before/during/after stills are from this same clip. This is a verified V2 dataset example, not an identified VLA-JEPA training example. Dataset:

<https://www.qualcomm.com/developer/software/something-something-v-2-dataset> . Mirror:

<https://huggingface.co/datasets/morpheushoc/something-something-v2> . Robot footage: real DROID episode AUTOLab+0d4edc83+2023-10-27-19h-52m-50s, exterior camera 24400334. Cropped; approximate timing restored from the 5.901-second recorded control-timestamp span, with an end hold. Paired trajectory.h5 contains recorded controls. Both videos play once silently, then hold. This is dataset footage, not a VLA-JEPA policy rollout. <https://droid-dataset.github.io/visualizer/>

Scene 3 - Noah

Feature prediction is shared; the pipeline differs

Clock 1:00-1:44; measured allocation 43.8 seconds.

0:00 - Begin conversationally; allow the example or shared objects to establish context.

0:17 - Stress the distinction between transition tokens and the future-state target.

0:35 - Settle on the output, then make the handoff without a long pause.

Dialogue

It's tempting to say that earlier methods predict pixels while JEPA predicts features. But that would miss an important detail. LAPA learns discrete latent actions using pixel reconstruction. UniVLA already reconstructs DINO features, conditioned on language, and then trains a policy to predict discrete action codes. VLA JEPA instead obtains continuous latent tokens directly from the policy's current observation and instruction. Those tokens help a world predictor estimate future state features, so the alignment loss trains the policy representation itself. The distinction is the information pathway and training pipeline, not simply pixels versus embeddings. Robot demonstrations then teach the action head how to turn that representation into control.

LAPA (2024), §3; UniVLA (RSS 2025), §III; VLA-JEPA (2026), §3. Different targets and pipelines; feature prediction is not unique to VLA-JEPA.

Scene 4 - Noah

Predict a future state in feature space

Clock 1:44-2:29; measured allocation 45.2 seconds.

0:00 - Begin conversationally; allow the example or shared objects to establish context.

0:18 - Stress the distinction between transition tokens and the future-state target.

0:36 - Settle on the output, then make the handoff without a long pause.

Dialogue

Think of watching someone put a jar into a box. You can anticipate the outcome without drawing the next frame in your head. That's a useful analogy, although the model's features aren't literal labels like jar inside box. Here's the precise distinction: the latent action tokens represent the intended transition. The world predictor uses those tokens and encoded state history to predict the next state's embedding. A frozen V JEPA two encoder processes the actual future video to provide the target. We compare those two feature representations. We don't reconstruct a future image, and these features aren't robot joint coordinates. Robot control comes from a separate action head. Let's follow the same pieces through the model.

VLA-JEPA, §3.2. Human frames: SSV2 validation example #174198. Diagram features are illustrative, not measured embeddings or explicit robot coordinates.

Scene 5 - Noah

First, encode the observation and instruction

Clock 2:29-2:52; measured allocation 23.0 seconds.

0:00 - Focus on observation and instruction.

0:07 - The VLM processes both inputs.

0:15 - Stress “internal representation”; lower pitch on “not motor commands.”

Dialogue

First, the policy reads the current observation and the language instruction. Its vision language backbone is Qwen three V L, with two billion parameters. The output we're focusing on is a set of latent action tokens. Think of these as an internal description of a transition. They will help predict what changes, but they aren't yet commands we can send to a robot.

VLA-JEPA, §3.2. Human frames: SSV2 validation example #174198. Diagram features are illustrative, not measured embeddings or explicit robot coordinates.

Scene 6 - Noah

Predict the next state's embedding

Clock 2:52-3:15; measured allocation 23.0 seconds.

0:00 - Begin conversationally; allow the example or shared objects to establish context.

0:09 - Stress the distinction between transition tokens and the future-state target.

0:18 - Settle on the output, then make the handoff without a long pause.

Dialogue

Now carry those tokens into the world predictor. The frozen video encoder represents the observed state history. The predictor combines that history with the latent action tokens and estimates the next state's embedding. So the tokens describe a transition, while the output describes the future state. There's no image decoder here. The actual future supplies a training target; it doesn't enter the policy branch.

VLA-JEPA, §3.2. Human frames: SSV2 validation example #174198. Diagram features are illustrative, not measured embeddings or explicit robot coordinates.

Scene 7 - Noah

Compare the prediction with the observed future

Clock 3:15-3:41; measured allocation 25.9 seconds.

0:00 - Focus on the actual future and frozen encoder.

0:08 - Identify target and prediction by both labels and colors.

0:17 - Slow on “stays fixed”; explain that the prediction path learns.

Dialogue

Finally, we need something to check that prediction against. The same frozen encoder processes the actual future frames and produces the target embedding. The alignment loss compares the prediction with that target. Training updates the prediction path, including the vision language model, while the target encoder stays fixed. That's how video supervision shapes the policy's internal representation.

VLA-JEPA, §3.2. Human frames: SSV2 validation example #174198. Diagram features are illustrative, not measured embeddings or explicit robot coordinates.

Scene 8 - Noah

The predictor sees history, while future states are targets

Clock 3:41-4:22; measured allocation 41.0 seconds.

0:00 - Clearly separate policy inputs from world-model history.

0:13 - Slow on teacher forcing and causal attention.

0:27 - Emphasize the future is a comparison target, then hand off to Fernando.

Dialogue

There's an important boundary here: the policy doesn't get to see the answer. Its latent action tokens come from the initial observations and the instruction. The world model can use state history during training, through teacher forcing, but its attention stays causal across time. In other words, it can use earlier states, not the future state it's supposed to predict. The future embeddings sit on the target side of the comparison. That separation matters because otherwise a good prediction could just be a shortcut, without learning a useful transition. So far we've learned a representation. Fernando will explain how that becomes an action.

VLA-JEPA, §3.2. Human frames: SSV2 validation example #174198. Diagram features are illustrative, not measured embeddings or explicit robot coordinates.

Scene 9 - Fernando

Learn the velocity from one training pair

Clock 4:22-4:58; measured allocation 35.3 seconds.

0:00 - Show the blue demonstration and gray noise before introducing mixing.

0:08 - Follow the orange mixed sample, without reading symbols.

0:17 - Explain target direction, then the predicted arrow.

0:26 - As the predicted arrow aligns, stress “match”; qualify the 2D illustration.

Dialogue

Thanks, Noah. To understand the action head, let's start with one training example. The blue point represents a demonstrated action chunk. We also sample a noise point, shown in gray. During training, we choose a point somewhere between them. The orange arrow is the direction from that noise sample toward the demonstration. The network sees the mixed point, the mixing time, and the latent action tokens, and predicts a velocity. We train that prediction to match the orange arrow. Then we repeat this with many pairs and many mixing times. This drawing is a two dimensional illustration; the real points represent higher dimensional action chunks.

Paper §3.3, Eqs. 7-8. Mixing time is generation time, not physical robot time. The target velocity is demonstration minus sampled noise; squared-error regression. Own 2D illustration inspired by Jia-Bin Huang and Julia Turc videos; not an empirical trajectory.

Scene 10 - Fernando

Generate actions by following the learned field

Clock 4:58-5:27; measured allocation 29.4 seconds.

0:00 - Start with the distinction: no demonstration is provided at inference.

0:09 - Follow the successive updates; do not move the pointer continuously.

0:19 - Explain four steps as a paper setting; the bent path is illustrative.

Dialogue

Generating an action is a different process. This time we don't have a demonstration to aim at. We start with fresh noise and ask the learned velocity field which way to move. After each small update, we ask again at the new position. Following that field gradually gives us an action chunk, conditioned on the latent tokens. The paper uses four integration steps in its reported setup. The bend in this sketch is illustrative: the straight training pair doesn't mean every generated path has to be straight.

Paper §3.3 and Appendix A: four denoising/integration steps, 7-dimensional actions, future action horizon 7. Own illustrative trajectory; sampler type is not asserted. Training and inference paths are different concepts.

Scene 11 - Fernando

Human and robot data provide different supervision

Clock 5:27-6:03; measured allocation 35.9 seconds.

0:00 - Separate human-video prediction supervision from robot action supervision.

0:12 - Slow on the weighting of the two objectives.

0:23 - End on the role of robot data in physical control.

Dialogue

Now the two sources of training data fit together. Human videos provide the predictive alignment objective, because they tell us what happened next. Robot demonstrations provide that objective too, plus the action labels needed for flow matching. The robot loss combines the action objective with a weighted world model objective; beta controls that weighting. Pretraining uses roughly two hundred and twenty thousand human videos and seventy six thousand robot trajectories. The important distinction is that human video helps shape the representation, while robot data still anchors it to physical control.

Paper, equation 9, sections 3.3 and 4.1; SSv2 220K videos, DROID 76K trajectories. Equation 5 is presented as an embedding discrepancy without an explicit norm; this presentation does not invent an L1 or Smooth-L1 choice.

Scene 12 - Fernando

Standard LIBERO: a near tie

Clock 6:03-6:18; measured allocation 14.9 seconds.

0:00 - Read the two percentages naturally, then stress “basically a tie.”

0:07 - Keep the conclusion modest; do not celebrate 0.1 pp.

Dialogue

On standard LIBERO, the headline result is basically a tie: ninety seven point two percent for VLA JEPA, versus ninety seven point one for OpenVLA O F T. So this comparison alone isn't a strong argument for the method.

Paper, Tables 1 and 3. Differences: $97.2 - 97.1 = 0.1$ percentage points, $79.5 - 69.6 = 9.9$ percentage points. Selected baseline is OpenVLA-OFT; this is not a claim of superiority to every model on every task. Paper reports 50 episodes per task on standard LIBERO and does not provide confidence intervals for these headline averages.

Scene 13 - Fernando

LIBERO-Plus: a larger gain under perturbations

Clock 6:18-6:43; measured allocation 24.8 seconds.

0:00 - Name the perturbations before comparing numbers.

0:08 - Stress “nine point nine percentage points”, not percent improvement.

0:16 - Settle on the limit: an average hides remaining failures.

Dialogue

The more interesting comparison is LIBERO Plus, which changes things such as the camera, lighting, and object layout. Here VLA JEPA reaches seventy nine point five percent, compared with sixty nine point six for OpenVLA O F T. That's a gain of nine point nine percentage points. This supports the robustness argument under these particular shifts, although an average doesn't tell us which failures remain.

Paper, Tables 1 and 3. Differences: $97.2 - 97.1 = 0.1$ percentage points, $79.5 - 69.6 = 9.9$ percentage points. Selected baseline is OpenVLA-OFT; this is not a claim of superiority to every model on every task. Paper reports 50 episodes per task on standard LIBERO and does not provide confidence intervals for these headline averages.

Scene 14 - Fernando

Human video improves LIBERO-Plus robustness

Clock 6:43-7:04; measured allocation 21.1 seconds.

0:00 - Frame this as a within-method ablation.

0:07 - Stress the 16.6 percentage-point difference.

0:14 - Explain why this comparison isolates the human-video contribution more directly.

Dialogue

What happens if we remove the human video training? On LIBERO Plus, success drops from seventy nine point five to sixty two point nine percent. That's sixteen point six percentage points within the same method. This is stronger evidence for the contribution of human video than just comparing two different models, because the ablation changes that part of training directly.

Paper, Tables 1-3 and sections 4.4-4.5. Within-method ablation deltas with human video minus without: LIBERO-Plus +16.6 pp; LIBERO +1.1 pp; SimplerEnv Google -13.2 pp. Real-world repeated grasping is an author-reported qualitative observation, with a proposed attribution to human videos, not an isolated causal proof.

Scene 15 - Fernando

Human video does not improve every benchmark

Clock 7:04-7:23; measured allocation 19.2 seconds.

0:00 - Use a neutral pitch to introduce the counterexample.

0:09 - Stress the drop, then narrow the overall claim.

Dialogue

But the benefit isn't universal. On the Google robot tasks in SimplierEnv, adding human video reduces success from seventy eight point four to sixty five point two percent. So we should say the evidence is strongest for robustness on LIBERO Plus, rather than claim that human video improves every task.

Paper, Tables 1-3 and sections 4.4-4.5. Within-method ablation deltas with human video minus without: LIBERO-Plus +16.6 pp; LIBERO +1.1 pp; SimplierEnv Google -13.2 pp. Real-world repeated grasping is an author-reported qualitative observation, with a proposed attribution to human videos, not an isolated causal proof.

Scene 16 - Fernando

Single GPU Quantization Trade-offs

Clock 7:23-7:57; measured allocation 34.3 seconds.

0:00 - Explicitly identify these as our preliminary results.

0:11 - Contrast memory and retention at INT8.

0:22 - Slow on the INT4 accuracy loss and action drift; conclude cautiously.

Dialogue

We also have our own preliminary single GPU results. On an R T X thirty ninety, the full precision setup peaks at nineteen point four gigabytes. Eight bit quantization brings that down to eleven point eight, while retaining ninety six point eight percent of baseline success. That's roughly forty percent less memory. Four bit quantization reaches seven point two gigabytes, but task accuracy falls below sixty percent and we saw action drift. So eight bit looks like the more promising compromise in this pilot. These are our preliminary deployment findings, separate from the paper's benchmark results.

Presenters' original slides, "Single GPU Quantization Trade-offs," confirmed as preliminary runs. FP16/B16: 19.4 GB peak VRAM and 100% relative success; INT8/Q8: 11.8 GB and 96.8% baseline-success retention; INT4/Q4: 7.2 GB with severe action drift and <60% task accuracy. These performance labels are kept distinct. INT8 memory reduction is $(19.4 - 11.8) / 19.4 = 39.2\%$. RTX3090 has 24 GB. These are our preliminary results, not results of Sun et al. The original does not specify trial count, checkpoint or uncertainty. "Promising" avoids claiming a demonstrated optimum.

Scene 17 - Fernando

Deployment gaps remain beyond headline averages

Clock 7:57-8:26; measured allocation 28.5 seconds.

0:00 - Explain the sensor-noise weakness.

0:09 - Reveal the ten-trial evidence with the second spoken segment.

0:19 - Reveal training versus deployment with the third segment.

Dialogue

There are still gaps behind those averages. Under sensor noise, VLA JEPA trails pi zero in the reported comparison.

The real robot evaluation also has only ten trials per task, with some wrong object selections. That limits how much we can conclude about reliability.

And the paper trains on eight A one hundred GPUs. Our single GPU pilot is about deployment, so it doesn't establish that the same training is cheap.

Paper, Table 3; sections 4.1 and 4.4; Appendix B. Ten trials per task refers to the real-world study. Eight A100 GPUs describes paper training, not a minimum inference requirement. Our preliminary deployment results are sourced from the presenters' original slides.

Scene 18 - Fernando

Predictive supervision strengthens the policy

Clock 8:26-8:45; measured allocation 19.4 seconds.

0:00 - Deliver the takeaway with a falling pitch.

0:09 - Lift on the discussion question, then thank the audience.

Dialogue

The takeaway is that predicting future embeddings can give a robot policy useful supervision, with the clearest evidence here coming from robustness under perturbations. The next question we'd ask is: which experiment would separate visual robustness from actual physical reasoning? That's a useful starting point for discussion. Thank you.

Paper, sections 3 and 4.5. Suggested discussion experiment: controlled appearance changes versus changed dynamics, under matched data and compute. This is our proposed test, not a result in the paper.

Implementation reference

The reported world predictor has 12 layers, 2 camera views, and an 8-frame future horizon. The action head uses 7-dimensional actions with four denoising/integration steps. These are reproduction details; the main scenes focus on what information flows through the model and what training objective teaches it.